Materiały dydaktyczne przeznaczone wyłącznie dla studentów wydziału Informatyki i Telekomunikacji

Przesyłanie danych multimedialnych

MPEG-TS - Transport stream - Header

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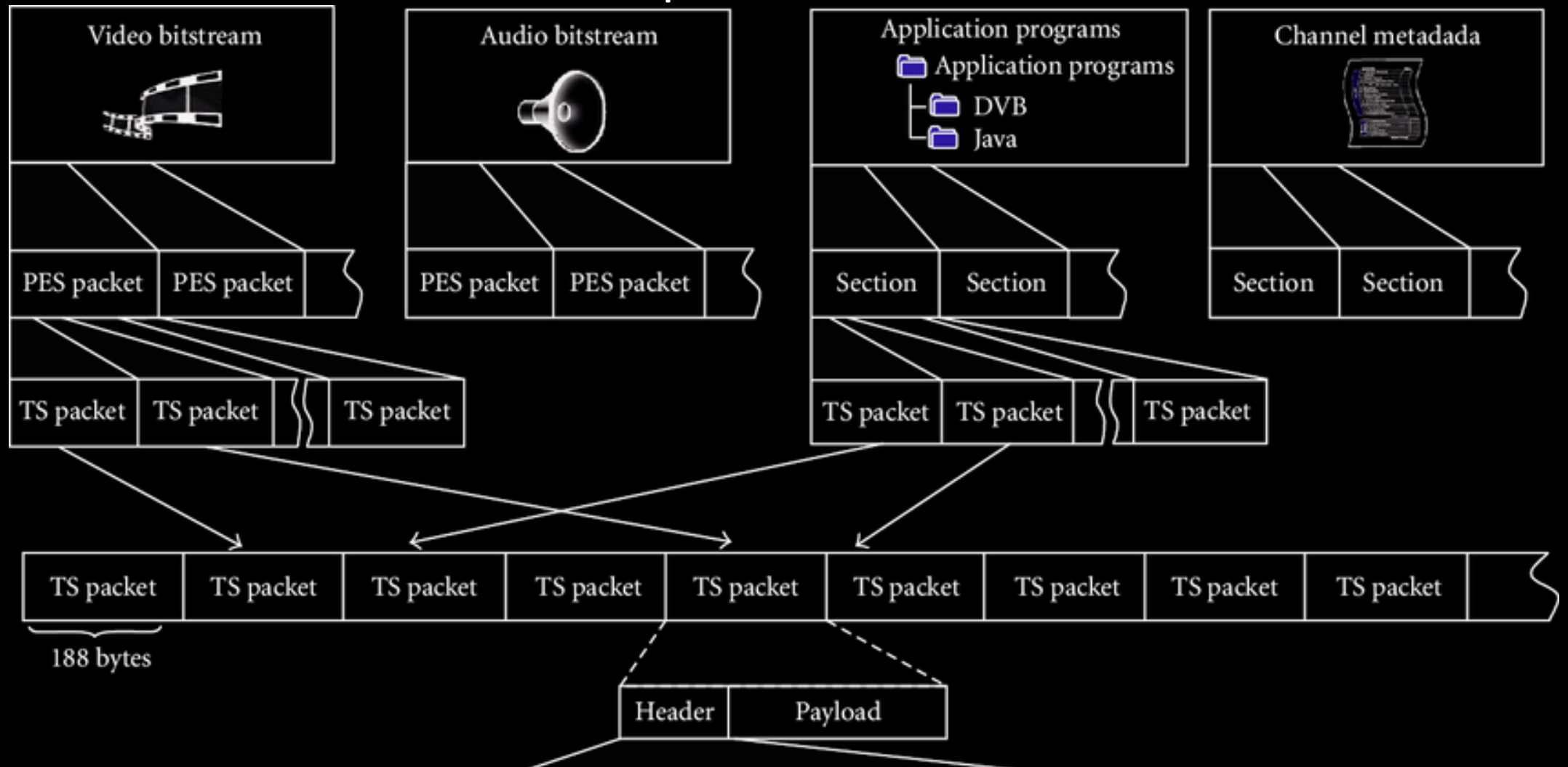
Wprowadzenie



Field	Sync. byte	Transport error indicator	Payload unit start indicator	Transport priority	Packet ID (PID)	Transport scrambling control	Adaptation field control	Continuity count
Number of bits	8	1	1	1	13	2	2	4

Wprowadzenie

```
Input #0, mpegts, from 'P1SATHD.ts':
  Duration: 00:00:42.17, start: 36062.437467, bitrate: 59977 kb/s
  Program 15702
    Metadata:
      service_name    : Discovery Life HD
      service_provider: nc+
    Stream #0:9[0xa1]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:10[0x54](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:11[0x55](eng): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:12[0x1f6](pol): Subtitle: dvb_teletext ([6][0][0][0] / 0x0006)
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15703
    Metadata:
      service_name    : VOX MUSIC TV
      service_provider: nc+
    Stream #0:5[0xa2]: Video: h264 (High) ([27][0][0][0] / 0x001B), yuv420p(tv, bt470bg, top first), 720x576
[SAR 16:11 DAR 20:11], 25 fps, 25 tbr, 90k tbn, 50 tbc
    Stream #0:6[0x58](pol): Audio: mp2 ([4][0][0][0] / 0x0004), 48000 Hz, stereo, s16p, 192 kb/s
    Stream #0:7[0x229]: Unknown: none ([5][0][0][0] / 0x0005)
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15704
    Metadata:
      service_name    : TVS
      service_provider: nc+
    Stream #0:14[0xa3]: Video: h264 (High) ([27][0][0][0] / 0x001B), yuv420p(tv, bt470bg, top first), 720x576
[SAR 16:11 DAR 20:11], 25 fps, 50 tbr, 90k tbn, 50 tbc
    Stream #0:30[0x5c](pol): Audio: mp2 ([4][0][0][0] / 0x0004), 48000 Hz, stereo, s16p, 192 kb/s
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15705
    Metadata:
      service_name    : TVP Kultura
      service_provider: nc+
    Stream #0:47[0xa4]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:48[0x60](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:49[0x61](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:25[0x1f9](pol): Subtitle: dvb_teletext ([6][0][0][0] / 0x0006)
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15706
    Metadata:
      service_name    : WP
      service_provider: nc+
    Stream #0:50[0xa5]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:51[0x64](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15707
    Metadata:
      service_name    : CBS Reality
      service_provider: nc+
    Stream #0:52[0xa6]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:53[0x68](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:54[0x69](eng): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15708
    Metadata:
      service_name    : TVP SERIALE
      service_provider: nc+
    Stream #0:55[0xa7]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:56[0x6c](pol): Audio: mp3 ([4][0][0][0] / 0x0004), 0 channels
    Stream #0:57[0x6d](pol): Audio: mp3 ([4][0][0][0] / 0x0004), 0 channels (visual impaired)
    Stream #0:20[0x1fc](pol): Subtitle: dvb_teletext ([6][0][0][0] / 0x0006)
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
  Program 15709
    Metadata:
      service_name    : AXN SPIN HD
      service_provider: nc+
    Stream #0:39[0xa8]: Video: h264 ([27][0][0][0] / 0x001B), none, 90k tbr, 90k tbn
    Stream #0:40[0x70](pol): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:41[0x71](eng): Audio: ac3 ([6][0][0][0] / 0x0006), 0 channels
    Stream #0:2[0x29a]: Unknown: none ([192][0][0][0] / 0x00C0)
    Stream #0:3[0x29b]: Unknown: none ([193][0][0][0] / 0x00C1)
……
```

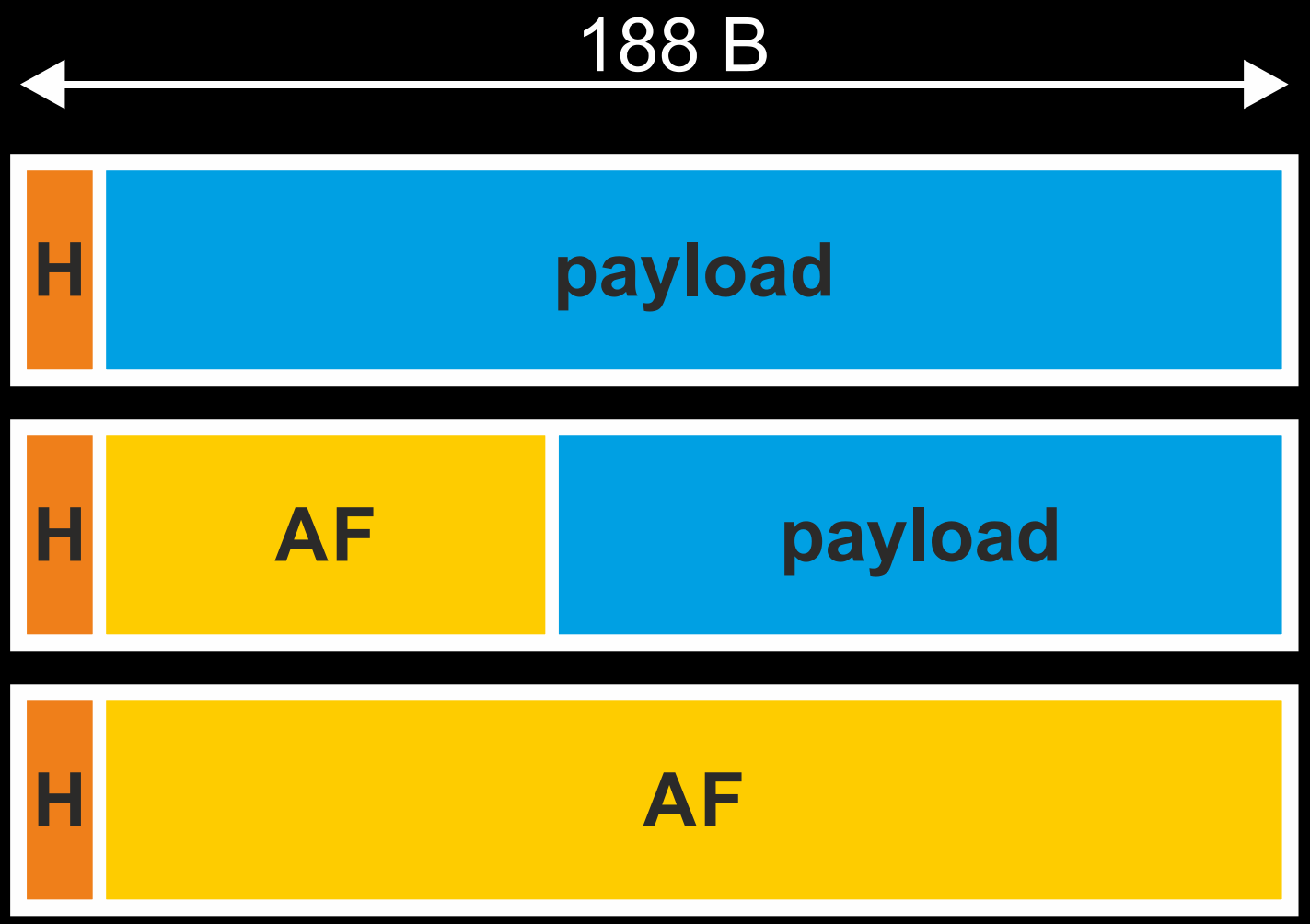
Wprowadzenie

- Strumień transportowy
  - pakiety o długości 188B (~4 ramki ATM)
  - kolejność bajtów – big-endian

- Warto poczytać:
  - en.wikipedia.org/wiki/MPEG_transport_stream
  - en.wikipedia.org/wiki/Endianness

Wprowadzenie

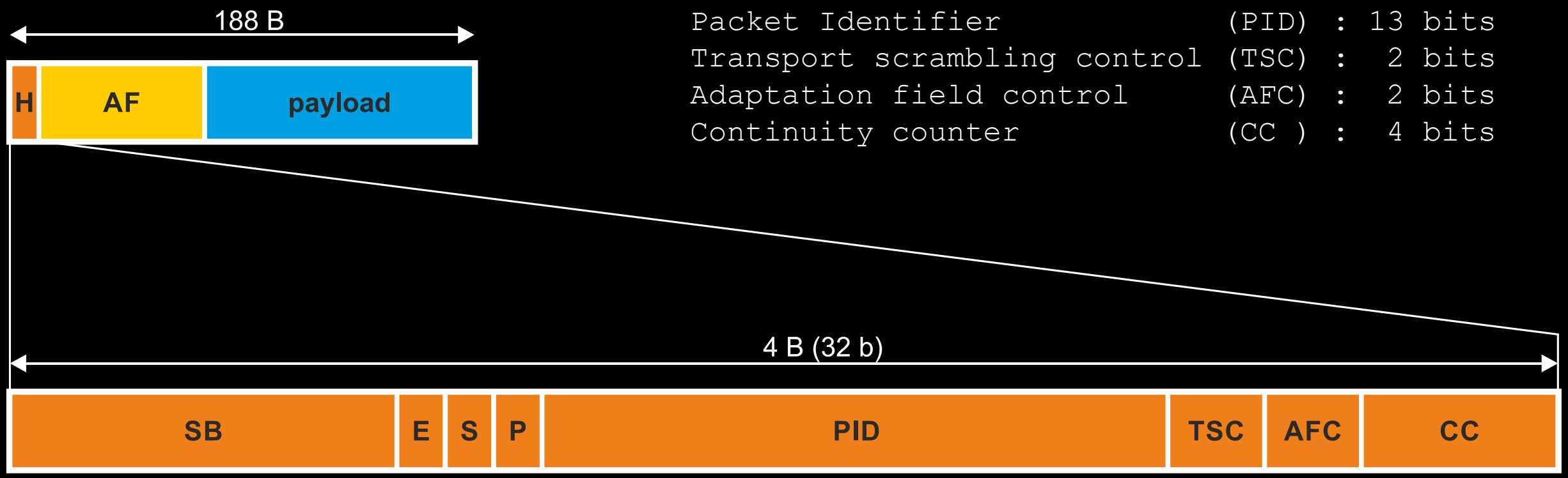
- Strumień transportowy – pakiet
  - H – header (nagłówek)
  - AF – Adaptation Field



Wprowadzenie

- Strumień transportowy – nagłowek pakietu
  - długość = 4B (32b)

```
Sync byte = 0x47 = `G`          (SB ) :  8 bits
Transport error indicator       (E  ) :  1 bit
Payload unit start indicator    (S  ) :  1 bit
Transport priority              (T  ) :  1 bit
Packet Identifier               (PID) : 13 bits
Transport scrambling control    (TSC) :  2 bits
Adaptation field control        (AFC) :  2 bits
Continuity counter              (CC ) :  4 bits
```



Informacje

- Potrzebne materiały:
  - Zestaw narzędzi: pakiet ffmpeg
    - `ffprobe`
    - `ffplay`
  - Strumień testowy:
    - `example_new.ts`

Analiza zawartości strumienia

- cmd: `ffprobe -i example_new.ts`

```
Input #0, mpegts, from 'example_new.ts':
  Duration: 00:10:00.54, start: 1.400000, bitrate: 2205 kb/s
  Program 1
    Metadata:
      service_name    : Service01
      service_provider: FFmpeg
    Stream #0:0[0xae]: Video: h264 (High) ([27][0][0][0] / 0x001B), yuv420p(tv, bt470bg, top
first), 720x576 [SAR 16:11 DAR 20:11], 25 fps, 50 tbr, 90k tbn, 50 tbc
    Stream #0:1[0x88](pol): Audio: mp2 ([3][0][0][0] / 0x0003), 48000 Hz, stereo, fltp, 192 kb/s
```

- Zawartość:
  - strumień wizji PID=0xAE(174), format=MPEG-4 AVC
  - strumień fonii PID=0x88(136), format=MPEG Audio / Layer 2

Parser strumienia transportowego

- Szablon projektu
- Krok 1:
  - załadowanie pakietu z pliku „.ts”
    - przydatne funkcje: `fopen`, `feof`, `fread`
    - plik otwarty w trybie binarnym
    - liczba wczytanych bajtów
  - parsowanie nagłówka pakietu strumienia transportowego
    - wszystkie pola
    - endianes
      - instrukcja BSWAP (www.felixcloutier.com/x86/bswap)
  - wyświetlenie zawartości pakietu
    - numer pakietu
    - wartości pól

Przykład

```
0000000000 TS: SB=71 E=0 S=1 P=0 PID=  17 TSC=0 AF=1 CC= 0
0000000001 TS: SB=71 E=0 S=1 P=0 PID=   0 TSC=0 AF=1 CC= 0
0000000002 TS: SB=71 E=0 S=1 P=0 PID=4096 TSC=0 AF=1 CC= 0
0000000003 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0
0000000004 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 1
0000000005 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 2
0000000006 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 3
0000000007 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 4
0000000008 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 5
0000000009 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 6
0000000010 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 7
0000000011 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 8
0000000012 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 9
0000000013 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=10
0000000014 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=11
0000000015 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=12
0000000016 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=13
0000000017 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=14
0000000018 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=3 CC=15
0000000019 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0
0000000020 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 1
0000000021 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 2
0000000022 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 3
0000000023 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 4
0000000024 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 5
0000000025 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 6
0000000026 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 7
0000000027 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 8
0000000028 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 9
0000000029 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=10
0000000030 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=11
0000000031 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=12
0000000032 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=13
0000000033 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=14
```